

Protocol for a Scoping Review of (Non)Collaboration in Network Interdiction Problems

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Abstract

Collaboration is a key strategy in fighting crime, and Network Interdiction Problems (NIP) are a common tool used for the same purpose. Some NIP research models multiple agents explicitly, but very few model the collaboration among them.

This scoping review will be reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) Statement. The main goal is to clarify key concepts and modeling strategies in the literature, the main assumptions regarding collaboration, and how those strategies affect the model's complexity. We will also identify the main gaps in the literature, which we hope can inspire new researchers to pursue that path.

We will conduct a comprehensive search in four databases, Engineering Village, Web of Science, Scopus, and Business Source Ultimate, with a set of keywords related to Collaboration or Multiple Agents in Network Interdiction Problems. We will exclude Editorials, Reports, White Papers, other grey literature, and papers written in languages other than English. A pilot screening will be conducted by the first author, with the goal of informing the later decisions on search strategy, potential themes, and categories for analysis. The protocol will be registered at OSF. The screening procedures and the evaluation will be conducted by both authors independently using the software Covidence, and contradictions will be debated until consensus is reached. Data will be charted in parallel by both authors in Covidence, using a set of data items on model overview, players, objective function, collaboration, and intervention. The synthesis will be descriptive, organizing the findings by main themes.

A critical appraisal of evidence and ethics approval will not be required, given the nature of the research topic: an overview of methodologies in Network Optimization.

The findings of this scoping review are intended for publication in peer-reviewed journals with an OR/IE readership.

Keywords: Collaboration; Multiple Agents; Network Interdiction; Scoping Review; bilevel optimization; interdiction games

1 Introduction

Since the 1960s, Network Interdiction Problems (NIP) have been used to help decision-making in anti-crime and national security issues Wollmer (1964). However, despite the recognized importance of collaboration in combating crimes, collaboration among agents is still an understudied issue when modeling NIP Cherniavskiy et al. (2019), Marsh (2023), Staniforth et al. (2019).

The first papers published on network disruptions were Wollmer (1964) and Fulkerson and Harding (1977). After a hiatus of important publications, we have seen a 'revival' in the 1990s with the first paper in network interdiction Wood (1993), followed by others like Cormican et al. (1998), Washburn and Wood (1995). Those papers usually deal with models to help decision-making on the screening of smuggling or other national security issues. The models that came after, especially from the 2010s forward, deal with a wider variety of topics under security, like cyber-security DuBois et al. (2023), Garnaev et al. (2014), drug trafficking Baycik (2018), Magliocca et al. (2019), and human trafficking Tezcan and Maass (2022), among others. There is a vast literature on the topic, and at least two major reviews were published in the 2020s Smith and Song (2020) and Anzoom et al. (2021). The latter focuses not only on NIP but also on other models to combat illicit supply chains. Both reviews indicate gaps in using collaborative aspects of anti-crime agents in modeling efforts. As of today, there are not many papers that deal with multiple agents collaborating or with the interdependence of multiple agents acting in a common network. The previous work known to the authors that motivated this review deals with decentralized agents Sreekumaran et al. (2015) or offers options for multiple agents acting on a common network Brown et al. (2011), He et al. (2018), Kosmas et al. (2024), Unsal (2010), Washburn and Wood (1995), Wilt (2019). Understanding how those papers approach collaboration, and the assumptions that limit their decisions, is crucial to see what can still be done in the field.

Collaboration among multiple agents is still under-researched, and to the best of our knowledge, no review like the one we are proposing has been published. Implementing collaborative aspects into NIP models can lead to more realistic and useful decision-making models that ultimately might thwart some illicit activities. A scoping review will be conducted to systematically map the research done in this area, as well as to identify existing gaps in knowledge. Our focus will be on synthesizing modeling strategies and modeling assumptions, and on offering potential guidelines for future research. We hope to offer a comprehensive overview of the topic, mapping the whole picture and synthesizing how research has been conducted on collaboration or non-collaboration of multiple agents in network interdiction problems. Shining light on this still under-researched topic and modeling strategy, so that it can inspire future researchers, is also a motivator.

1.1 *Some important definitions:*

Some definitions will guide our research efforts. Not all terms will be used as keywords or as inclusion or exclusion criteria, but they are necessary for understanding the topic. The words defined below don't always have homogeneous meanings throughout the Operations Research field, nor within the Network Optimization subfield. To mitigate potential ambiguities, we define the meanings we are using as follows:

- **Stackelberg game and Network interdiction problems:** The Stackelberg Game is a type of game where 'leaders' are capable of committing to their actions before other players, the 'followers.' Those are players that must respond to the leaders' decisions. In the Network Interdiction Problem (NIP), we usually have a Stackelberg type of game where we are dealing with a set of opposing players, where the leader takes interdiction actions to *max-min* or *min-max* the objective function that a follower can obtain in solving its optimization problem. The leader's interdiction actions can impact the follower's objective or feasible region Smith and Song (2020). In NIP, we can also have Non-Zero-Sum Games Goldberg (2017), Okamoto et al. (2012), which means that the opposing players don't have symmetric objectives.
- **Opposing players:** In NIP, there is a set of players, also called "agents", that has an objective that directly conflicts with the objective of another set of players. For this paper, that is what we are calling "opposing players." They might have directly opposite goals (zero-sum games) or compete with different objective functions. The opposing sets of players are the decision-makers for each of the two or more levels of decision in the model. There are different names that define opposing players in the NIP field. More commonly, they are called defenders and attackers; interdictors and operators; or disruptors and followers, among others. Throughout this paper, we are calling the opposing players attackers (outer problem) and operators (inner problem).
- **Bi-level or Multi-level:** A Bi-level problem is a way to model the opposing players' decision problem. The first level is the leader, and the second (or more) is the other set(s) of players. They are usually modeled as an outer and an inner problem, where the outer problem contains the inner problem as a constraint. The outer problem represents the perspective of the attacker. This set of players aims to degrade the network's performance by strategically modifying certain elements (like removing edges, reducing capacities, or increasing costs). The inner problem, on the other hand, represents the perspective of the network operator or defender. After the interdiction has made its moves, the inner problem involves optimizing the network's performance under these new conditions. This could involve finding the maximum flow or the shortest path, minimizing costs, or others Smith and Song (2020). For this review, we will

focus on problems where intervention decisions are in the outer problem, which aligns with the definition of a network interdiction problem.

- **two-stage or multi-stage NIP:** While the term "multi-stage" in Network Interdiction often refers to problems with multiple periods, it can also describe the bi-level structure mentioned above. In that definition, each stage is a period where actions can be taken. We are using the former as an inclusion criterion, but we are not excluding papers where multi-stage means multiple periods (the latter definition). In this paper, we are reviewing only bi-level or multi-level network interdiction in an s-t network.
- **network disruption:** Disruptions mean any action, random or not, that will degrade the structure, the parameters, or the decision of a network optimization problem Smith and Song (2020). For our purpose, network disruption will only be considered if it is caused by one of the decision-makers in the model. It might be affected by random elements, but it is always a consequence of an outer-problem player's decision.
- **network interdiction:** It represents the action one of the (sets of) players can take to disrupt the network in order to worsen the decision of the opposing agent. It can also take the name of intervention for our purpose. There is the possibility of players taking one or more interventions in the whole network or in a specific arc/edge/node. They might be of one or multiple types. We expect that each different intervention will have a heterogeneous effect on the problem, which is why they are modeled separately. When the model has heterogeneous intervention types, they will be called multiple interdiction. There might be some element of the network parameters or scenarios that is subject to randomness and is not deterministic. This will not be used as an exclusion criterion, but it will be part of the evaluation.
- **multiple players:** When there is more than one agent in the same group of "opposing players", the model is a multiple-agents model. The agents in the same group have the characteristic of having harmonious goals among them. The model might consider collaborative or non-collaborative relationships between these multi-agents (their "peer players", as we are defining for this paper). They might have full or incomplete knowledge of the decisions of their peers. Depending on the type of information they share, they can make that decision anywhere from totally centralized (one decides for all) to decentralized (each agent decides without knowledge of the other). They might act on separate subsets of the network or on the whole network. In this review, we are considering only papers that model the multiple peer players making their decisions as the outer problem.
- **centralized or decentralized decision-making:** Multiple players might make the decision together, centralized, or without knowledge of how their peers are making their decisions, decentralized. A centralized decision

means that players have total awareness of their peers' decisions. It is not an exclusion criterion for this review.

- **homogeneous or heterogeneous multi-agents:** The peer players might have no difference from each other, or the effect of their decisions might have different components or effects. We are going to evaluate either case.
- **asymmetric or symmetric information:** Among the opposing players or between peer players, agents might have limited information regarding any characteristic of the network (parameters, structure) or the decision of the other player, while others have full information. That is what is going to be considered asymmetry of information. In the same way, we are also considering symmetric or asymmetric information among opposing and peer players.
- **Collaboration:** Collaboration, for this paper, has two different evaluation components. First, it regards the knowledge each player has regarding their peers' actions. We might consider collaboration when peer players have a degree of intelligence on their peers' decisions when committing to their own decisions. This type is going to be called "information-sharing collaboration". The second considers the knowledge among peer agents about the interdependence of their decisions. If peer players commit to their actions considering the cumulative effects of combined decisions, that will be defined as "acknowledgment collaboration." They might not necessarily be aware of their peers' final decisions, but they know how their combined decision improves or damages the final network. In the review, we will classify each paper regarding the level of collaboration and the type of effect of their (lack of) collaboration. We will not use collaboration as an inclusion/exclusion criterion, but we are evaluating the collaborative or non-collaborative efforts.

1.2 Objectives

The primary goal of this scoping review is to assess the landscape of NIP models with multiple agents whose interactions are modeled in a way that might affect the outcome or the agents' final decision. We want to know the common modeling strategies and the main assumptions made. We also hope to identify gaps in the literature in the Network Optimization field regarding collaboration efforts. The scoping review offers a structured search procedure that helps identify the gaps and the practices that guide us to better and more realistic models Munn et al. (2018). We are not going to analyze the body of evidence related to Reinforcement Learning or Artificial Intelligence, nor any paper in the realm of Network Science, which studies patterns of connectivity among actors or things as defined by Barabási and Pósfai (2016).

1.2.1 Research Questions:

1. What are the modeling decisions made regarding the inclusion of multiple agents in NIP? How do those decisions change the model's objective function, the relationship between peer players' decisions on collaboration and interventions, and the uncertainty or asymmetry of information?
2. What are the current gaps in the literature for NIP with multiple peer players?
3. What are the assumptions used for modeling collaboration among those players in NIP models?
4. What are the solution strategies used to handle the added complexity?

2 Methods

Our protocol is drafted based on the PRISMA Extension for Scoping Reviews (PRISMA-ScR) reporting guidelines as defined by Tricco et al. (2018) and the methodology explained in the JBI Manual for Evidence Synthesis Peters et al. (2024). The final manuscript will be reported following PRISMA-ScR Tricco et al. (2018).

The protocol will be registered with the Open Science Framework (<https://osf.io/>) on 04 December 2024, and it will be reviewed by Northeastern University librarians. The protocol and the registration will be shared with all members of the team.

The screening of the Title, Keywords, and Abstract will be conducted by both authors independently using the software Covidence [Veritas Health Innovation, Melbourne, Australia]. All discrepancies or conflicts in inclusion/exclusion will be debated until a consensus is reached by the team members. If the abstract is not available, the record will be passed on automatically to full-text screening. When it is unclear whether the paper is within the inclusion criteria, the team member will assign a "maybe" tag, and the paper will be included in the pool for full-text evaluation. The screening of the full text and the evaluation process will be conducted by both authors independently using the same software. If a paper is tagged as "maybe" by two team members after the full-text review, its inclusion will be discussed by all team members until a consensus is reached. When an ambiguity arises that is not mentioned above, we will decide the procedure as a team and add it to the protocol.

A critical appraisal and ethics approval will not be required, given the nature of the research topic: an overview of methodologies in the Network Optimization field. The findings of this scoping review are intended for publication in peer-reviewed journals with an OR/IE readership.

Inclusion/exclusion criteria: Because our goal is to understand how collaboration or non-collaboration of multiple peer players has historically been

modeled using NIP models, eligible studies must be optimization models formulated as Network Interdiction Problems as defined in Section 1. They must also include multiple peer interdicting agents in the first-level problem, and the interdiction decisions those peer players make must somehow affect the outcome of the problem. We are including bi-level and multi-level models in the Network Optimization field. They must involve two or more peer players making their decisions in the outer problem, on a common s-t structured network. The intervention must be a decision variable of the model; studies where it is not will be excluded. Studies that do not fit this conceptual framework, including those where the multiple peer players are in the inner problem, will not be considered.

Eligible document types are peer-reviewed journal articles, preprints, book chapters, dissertations, and published review papers. We are excluding Editorials, Reports, White Papers, and other grey literature. Records whose only available text is a short abstract will also be excluded: when an abstract is relevant, we will keep it for the second round in order to determine whether it is an extended or a short abstract. We are considering studies with a publishing date up to the date of data collection, so we are not excluding studies by their publishing date. Clear exclusions are studies of Artificial Intelligence models (e.g., Machine Learning – ML or Reinforcement Learning – RL) and Network Science models. Those models are out of the scope of this review because they are not network interdiction problems, even though they share many of our keywords. Studies that were not published in English will also be excluded, English being the language both reviewers read fluently.

Information sources: Four databases were selected to perform the search, each through the interface named: Inspec Archives and Ei Compendex through Engineering Village, Web of Science, Scopus, and Business Source Ultimate through EBSCO. Between them, these hold most of the literature in engineering, management science, and operations research, which are the fields where network interdiction models are published. The choice was made on the experience of the authors together with the university librarians who advised on the review. Preprints enter through Inspec Archives, which indexes arXiv, so the search reaches preprint versions without a separate grey literature strategy.

Two sources were considered and dropped. ProQuest will not be searched, because Engineering Village indexes doctoral dissertations, so only master’s theses would be added, which the librarians did not consider a significant limitation. Google Scholar will not be searched either, also on their advice, because its lack of reproducibility, its limits on bulk citation export, and its handling of Boolean operators make it unsuitable for evidence synthesis. The Defense Technical Information Center was named in an early draft of this protocol and then dropped from the final set. Together, the four databases index all of the papers and research previously known to us Brown et al. (2011), He et al. (2018), Kosmas et al. (2024), Sreekumaran et al. (2015), Washburn and Wood (1995), Wilt (2019) but one master thesis Unsal (2010), which is indexed only in the Defense Technical Information Center and so will not be reachable by this search.

The search will be executed by both authors, with Scopus searched by the second author, whose institutional access covers it. We will not add sources manually, so no citation searching or grey literature searching will supplement the database results. We will, however, contact authors to obtain or confirm model details that are not reported in the text.

2.1 *Search Strategy*

The first author leads the development and execution of the search strategy and the literature identification. We will follow the Peer Review of Electronic Search Strategies (PRESS) guideline to formulate the search strategy in the following steps McGowan et al.: (1) Translation of the research question; (2) Boolean and proximity operators; (3) Subject headings (database-specific); (4) Text word search; (5) Spelling, syntax, and line numbers; and (6) Limits and filters. In addition, a specialist librarian and the team members will review the search strategy.

The strategy is developed and validated in three moves. First, a pilot on Engineering Village alone, where most of the known sources are indexed, described in Section 2.2. Second, a limited search on October 21, 2024, again on Engineering Village, after which we analyzed the words appearing in titles and abstracts and the index terms used to describe the retrieved records, which let us curate the term blocks. Third, a validation procedure, in which we check whether the resulting strategy returns the studies already known to the authors as relevant, and report any that it misses.

The final search, using all the identified relevant keywords and index terms (Figure 1), will then be undertaken across all four included databases. Only studies published in English will be considered for inclusion in this review.

2.2 *Selection of sources of evidence*

We are selecting quantitative research focused on modeling strategy in the Engineering and Business fields, specifically in the Operations Research knowledge base. There will be two rounds of screening by the reviewers: (1) Title and Abstract screening; and (2) Full-text eligibility analysis.

A pilot is undertaken to test the inclusion/exclusion criteria and the data items to be collected, as well as the disambiguation and conflict rules. This pilot was conducted on August 21, 2024, by the first author, and its results will be discussed among the team members. The limited search for the pilot consisted of searching a single database, Engineering Village, where most of the known papers are indexed, for the keyword combination *Collaboration AND "Network Interdiction Problem"*, our main focus, in the Subject, Title, and Abstract fields. Eleven papers were found using those criteria. If conflicts are many, we will reevaluate the terms and the data charting strategy as a new pilot.

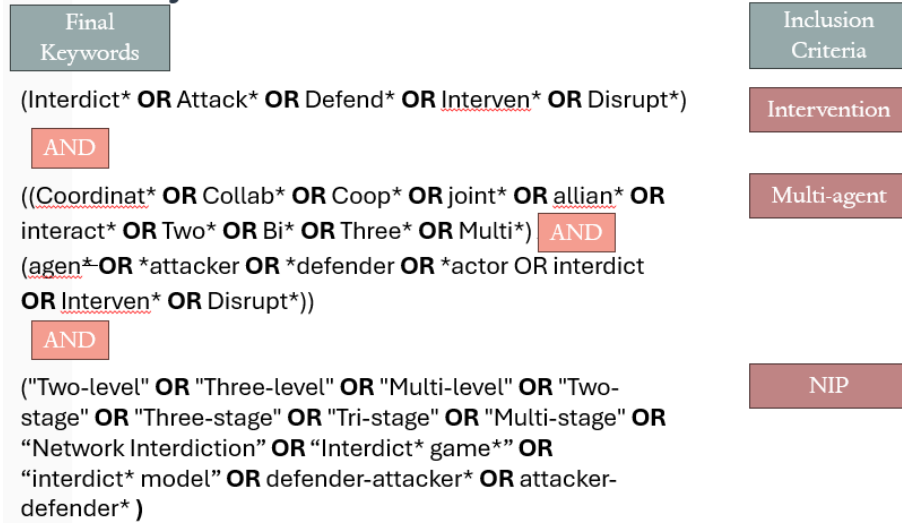


Figure 1: Final Keywords Combinations - Protocol

The Covidence software for managing systematic and scoping literature reviews will be used to remove duplicates, to include and exclude records (both authors), and to run the screening and extraction process for analysis.

2.3 Data charting process

The records will be stored in the Covidence software. Data charting will be done in parallel by the two independent reviewers (the authors), using Covidence as the tool. The study team developed a set of data items (Section 2.4) regarding assumptions, modeling decisions, and missing strategies in the literature on Collaborative Multi-agent NIP. The set of data items will be tested in the pilot before it is used. When a model detail we need is not reported in the text, we will contact the authors to obtain and confirm it.

2.4 Data items

The authors will collect data from the remaining articles using Covidence. This will be an iterative process of systematically extracting data items from the literature. The data collected will be limited by the research questions and the review objectives. The data items that will be extracted during this process can be seen in Table 1. This is a preliminary list that might change as the screening and collection process progresses, and weekly meetings among the team members will guide any change. Our focus is to analyze how multiple peer players might affect the network in a way that is different from the case where only one agent interdicts the network.

Theme	Data Items Extracted for the Scoping Review	Definition
General	Article details	Title, authors, year published, funding source, journal, type of paper (e.g.: peer-reviewed, pre-print)
	Optimization or Game Theory focus	Is the focus of the analysis on optimization or on game-theoretic methods?
	Application Context	What are attackers/operators meant to represent in the real world? And the network structure?
	Opposing Players definition	How are opposing players defined in the paper (e.g.: attackers x defenders)?
	One or Multiple Stages	Are decisions made in multiple periods of time or multiple stages?
Players	Solving Strategy	How do the authors deal with the complexity of the problem?
	Asymmetric information between Opposing Players	Is asymmetry in information considered for opposing players?
	Asymmetric information among Peer Players	Is asymmetry in information considered among peer players?
	Homogeneous or Heterogeneous peers	Are the peer agents affecting the network similarly or differently?
	Opposing decisions order	What is the decision order among opposing players?
Objective Function	Resources	Are the resources shared by Peer Players?
	Objective Function type	What is the problem each opposing player is optimizing (e.g.: Maximum Flow, Shortest Path...)?
	Difference on the OF for Opposing Players	Are the Objective Functions the same but opposite (Zero Sum), or different among opposing players?
	Difference on the OF for Peer Players	Are the Objective Functions the same or different among the peer players?
	Total, Partial, or Non Collaboration	What is the type of collaboration modeled?
Collaboration	Levels of collaboration	Are there different levels of collaboration considered?
	How Collaboration is defined	Is collaboration, if there is any, of the "information-sharing" or the "acknowledgment" type (Section 1)?
	Assumptions on Collaboration	What are the assumptions regarding collaboration?
	Centralized or Decentralized decision	Are the decisions made centralized or decentralized (Section 1)?
	Collaboration decision order	What is the decision order among Peer Players?
Intervention	Collaboration effects	How does collaboration affect the intervention effect?
	Intervention effect	What is affected by the decision of intervention?
	Intervention stochasticity	Are there any random elements in the formulation that affect parameters, structure, or decisions with respect to interventions?
	Collaboration stochasticity	Is the collaboration effect stochastic or deterministic?
	Disruption levels	Are there different levels of disruption when an intervention is made? How?
	Multiple interdictions per arc	Can multiple interdictions act on the same arc?

Table 1: Data Items Extraction for the Scoping Review

2.5 Synthesis of results

The PRISMA flow diagram will be generated automatically by Covidence, and it will be used to graphically illustrate the flow of information throughout the screening process. Our intended analysis is descriptive statistics using Microsoft Excel [Microsoft Corporation, Redmond WA, USA]. To synthesize the data found, we will organize the most relevant findings in a table by main themes (Table 2), together with graphs for the quantitative analysis (e.g.: number of papers per year, number of papers per publisher group, number of papers per main theme). Graphs will be added to explain how the themes distribute among the literature.

Themes	Citations
Theme 1: Collaboration as information sharing or as acknowledgment of interdependence	citations
Theme 2: Collaboration as multiple intervention types	citations
Theme 3: Solving Strategies	citations
Theme 4: Collaboration - main assumptions	citations
Theme 5: Application Contexts	citations

Table 2: Study Characteristics of Collaboration on Multiple Agents Network Interdiction Problems

We expect that this approach may change as the reviewing process progresses. Expert librarians will be invited in this phase to review the synthesis and charting procedures.

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A PRISMA-ScR checklist

Where each PRISMA-ScR item Tricco et al. (2018) is addressed in this protocol. Items 14 to 21 concern results and discussion and are not applicable at the protocol stage.

Item	Section	Addressed in
1	Title	Title
2	Abstract	Structured summary
3	Introduction	Section 1
4	Introduction	Objectives and Research Questions
5	Methods	Methods, first paragraphs (registration)
6	Methods	Inclusion/exclusion criteria
7	Methods	Information sources
8	Methods	Search Strategy; full strings in Appendix B
9	Methods	Selection of sources of evidence
10	Methods	Data charting process
11	Methods	Data items, Table 1
12	Methods	Critical appraisal not required (rationale given in Methods)
13	Methods	Synthesis of results
14–21	Results, Discussion	Not applicable at the protocol stage
22	Funding	Funding

Table 3: PRISMA-ScR items and where they are addressed in this protocol

The PRISMA-ScR checklist Tricco et al. (2018) follows.

Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) Checklist

A PRISMA-ScR CHECKLIST

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SECTION	ITEM	PRISMA-ScR CHECKLIST ITEM	REPORTED ON PAGE #
TITLE			
Title	1	Identify the report as a scoping review.	
ABSTRACT			
Structured summary	2	Provide a structured summary that includes (as applicable): background, objectives, eligibility criteria, sources of evidence, charting methods, results, and conclusions that relate to the review questions and objectives.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known. Explain why the review questions/objectives lend themselves to a scoping review approach.	
Objectives	4	Provide an explicit statement of the questions and objectives being addressed with reference to their key elements (e.g., population or participants, concepts, and context) or other relevant key elements used to conceptualize the review questions and/or objectives.	
METHODS			
Protocol and registration	5	Indicate whether a review protocol exists; state if and where it can be accessed (e.g., a Web address); and if available, provide registration information, including the registration number.	
Eligibility criteria	6	Specify characteristics of the sources of evidence used as eligibility criteria (e.g., years considered, language, and publication status), and provide a rationale.	
Information sources*	7	Describe all information sources in the search (e.g., databases with dates of coverage and contact with authors to identify additional sources), as well as the date the most recent search was executed.	
Search	8	Present the full electronic search strategy for at least 1 database, including any limits used, such that it could be repeated.	
Selection of sources of evidence†	9	State the process for selecting sources of evidence (i.e., screening and eligibility) included in the scoping review.	
Data charting process‡	10	Describe the methods of charting data from the included sources of evidence (e.g., calibrated forms or forms that have been tested by the team before their use, and whether data charting was done independently or in duplicate) and any processes for obtaining and confirming data from investigators.	
Data items	11	List and define all variables for which data were sought and any assumptions and simplifications made.	
Critical appraisal of individual sources of evidence§	12	If done, provide a rationale for conducting a critical appraisal of included sources of evidence; describe the methods used and how this information was used in any data synthesis (if appropriate).	
Synthesis of results	13	Describe the methods of handling and summarizing the data that were charted.	

SECTION	ITEM	PRISMA-ScR CHECKLIST ITEM	REPORTED ON PAGE #
RESULTS A PRISMA-ScR CHECKLIST			16
Selection of sources of evidence	14	Give numbers of sources of evidence screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally using a flow diagram.	
Characteristics of sources of evidence	15	For each source of evidence, present characteristics for which data were charted and provide the citations.	
Critical appraisal within sources of evidence	16	If done, present data on critical appraisal of included sources of evidence (see item 12).	
Results of individual sources of evidence	17	For each included source of evidence, present the relevant data that were charted that relate to the review questions and objectives.	
Synthesis of results	18	Summarize and/or present the charting results as they relate to the review questions and objectives.	
DISCUSSION			
Summary of evidence	19	Summarize the main results (including an overview of concepts, themes, and types of evidence available), link to the review questions and objectives, and consider the relevance to key groups.	
Limitations	20	Discuss the limitations of the scoping review process.	
Conclusions	21	Provide a general interpretation of the results with respect to the review questions and objectives, as well as potential implications and/or next steps.	
FUNDING			
Funding	22	Describe sources of funding for the included sources of evidence, as well as sources of funding for the scoping review. Describe the role of the funders of the scoping review.	

JB1 = Joanna Briggs Institute; PRISMA-ScR = Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews.

* Where *sources of evidence* (see second footnote) are compiled from, such as bibliographic databases, social media platforms, and Web sites.

† A more inclusive/heterogeneous term used to account for the different types of evidence or data sources (e.g., quantitative and/or qualitative research, expert opinion, and policy documents) that may be eligible in a scoping review as opposed to only studies. This is not to be confused with *information sources* (see first footnote).

‡ The frameworks by Arksey and O'Malley (6) and Levac and colleagues (7) and the JBI guidance (4, 5) refer to the process of data extraction in a scoping review as data charting.

§ The process of systematically examining research evidence to assess its validity, results, and relevance before using it to inform a decision. This term is used for items 12 and 19 instead of "risk of bias" (which is more applicable to systematic reviews of interventions) to include and acknowledge the various sources of evidence that may be used in a scoping review (e.g., quantitative and/or qualitative research, expert opinion, and policy document).

From: Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. *Ann Intern Med*. 2018;169:467–473. doi: 10.7326/M18-0850.

B Full search strategy

Keyword combination (as in Figure 1):

```
(Interdict* OR Attack* OR Defend* OR Intervene* OR Disrupt*) AND ((Coordinat*
OR Collab* OR Coop* OR joint* OR allian* OR interact* OR Two* OR Bi* OR
Three* OR Multi*) AND (agen* OR *attacker OR *defender OR *actor OR inter-
dict*)) AND ("Two-level" OR "Three-level" OR "Multi-level" OR "Two-stage" OR
"Three-stage" OR "Tri-stage" OR "Multi-stage" OR "Network Interdiction" OR "In-
terdict* game*" OR "interdict* model" OR defender-attacker* OR attacker-defender*)
```

Scopus (via Scopus; 1,569 records).** Run with the interface's default field set, which could not be specified exactly; all retrieved records were carried into screening.

```
( TITLE-ABS-KEY ( ( ( two-level* OR "two levels" OR three-level* OR
"three levels" OR bi-level* OR "bi level" OR bilevel* OR multilevel*
OR multi-level* OR "multi levels" OR interdict* OR defender-attacker*
OR attacker-defender* )
AND ( game* OR model* OR problem* OR network* OR "integer programming"
OR "linear programming" OR "mixed integer linear programming" OR programming ) ) )
AND TITLE-ABS-KEY ( ( ( two* OR three* OR multi* OR collaborati*
OR cooperati* OR coordinati*
OR joint* OR allian* OR interact* )
AND ( agen* OR player* OR attacker* OR defender* OR interdict* OR disrupt* ) ) )
AND TITLE-ABS-KEY ( ( attack* OR defen* OR interdict* OR intervent* OR disrupt* ) )
AND TITLE-ABS-KEY ( ( network ) ) AND LANGUAGE ( english ) )
```

Web of Science (Topic search - 1,193 records)

```
(( ( (TS=((two-level* OR "two levels" OR three-level* OR "three levels"
OR bi-level* OR "bi level" OR bilevel* OR multilevel* OR multi-level*
OR "multi levels" OR interdict* OR defender-attacker*
OR attacker-defender*))
AND (game* OR model* OR problem* OR network* OR "integer programming"
OR "linear programming" OR "mixed integer linear programming"
OR programming)))
AND TS=((two* OR three* OR multi* OR collaborati* OR cooperati*
OR coordinati* OR joint* OR allian* OR interact*))
AND (agen* OR player* OR attacker* OR defender* OR interdict*
OR disrupt*))
AND TS=(attack* OR defen* OR interdict* OR intervent* OR disrupt*))
AND TS=(network) ) AND LA=(English)
```

Business Source Ultimate (via EBSCO; 296 records). Run with the interface's default field set, which could not be specified exactly; all retrieved

records were carried into screening.

```
( ((two-level* OR "two levels" OR three-level* OR "three levels"
   OR bi-level* OR "bi level" OR bilevel* OR multilevel* OR multi-level*
   OR "multi levels" OR interdict* OR defender-attacker*
   OR attacker-defender*)
  AND (game* OR model* OR problem* OR network* OR "integer programming"
       OR "linear programming" OR "mixed integer linear programming"
       OR programming))
  AND ((Two* OR Three* OR Multi* OR Collaborati* OR Cooperati*
       OR Coordinati* OR joint* OR allian* OR interact*)
    AND (agen* OR player* OR attacker* OR defender* OR interdict*
        OR disrupt*))
    AND (attack* OR defen* OR interdict* OR intervent* OR disrupt*)
    AND (network) )
AND LA English
```